Date: 18 JUL 2024 **Regd. No.** | | | | | | | | | | |

Sub Code:23CS/IT/AD/AM2T04 **MIC23**

B. Tech II Semester Regular Examinations, July-2024

DATA STRUCTURES

(Common to CSE, IT, AI&DS, AI&ML)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A	**10 X 2 = 20 M**		
			Marks	**BTL**	**CO1**
I.	**a)**	Apply Bubble sort on the given elements: 5, 2, 9, 1, 8	2 M	L3	CO5
	b)	List out the collision resolution methods in Hashing?	2 M	L2	CO4
	c)	What is Queue overflow condition and when to apply this condition	2 M	L2	CO1
	d)	Give the time complexities of binary search and linear search	2 M	L1	CO1
	e)	What is the role of Stack in evaluating postfix expression	2 M	L2	CO3
	f)	Let the following queue can accommodate maximum six elements with the following data. front = 2, rear = 4 queue = __, __, L, M, N, __ How front, rear and queue have been updated after enqueue(P) operation takes place?	2 M	L4	CO4
	g)	5 / \\ 3 7 / \\ \\ 2 4 8 What is the result of performing a post-order traversal on the above Binary Search Tree?	2 M	L3	CO5
	h)	What is the result after evaluating the postfix expression 5, 4, 6, +, *, 4, 9, 3, /, +, * .	2 M	L3	CO3
	i)	How does accessing an element in an array differ from accessing an element in a linked list?	2 M	L3	CO2
	j)	Differentiate Single linked list and Double linked list	2 M	L3	CO2
		PART – B	**5 X 10 = 50 M**		
		UNIT – I			
1.	**a)**	What do you mean by space complexity and time complexity of an algorithm? Write an algorithm/pseudo code for linear search and mention the best case and worst case time complexity of Linear Search algorithm?	5 M	L2	CO1
	b)	Illustrate the working of Insertion sort on the following array with 7 elements : 30,45,25,32,55,60,49	5 M	L3	CO1
		(OR)			
2.	**a)**	Why Binary Search algorithm is more efficient that linear search? Depict your answer with suitable example? Mention the time complexity of two algorithms.	5 M	L3	CO1

	b)	Explain Merge Sort algorithm/pseudocode with the help of an example? Mention the best case and worst case time complexity of Merge sort algorithm?	5 M	L2	CO1
UNIT – II					
3.	a)	Write an algorithm/pseudocode to delete a node at the end of a doubly linked list.	5 M	L2	CO2
	b)	List various operations of linked list and explain how to insert a node in specified position of the list	5 M	L2	CO2
(OR)					
4.		Explain the structure for single linked list. Write an algorithm/function to insert an element at the beginning, at the end and at the given position.	10 M	L2	CO2
UNIT – III					
5.		Convert the given infix expression A+B^C+(D*E/F)*G into its postfix expression, and evaluate the same using stack. Here A=3, B=5, C=2, D=7, E=4, F=1, G=8.	10 M	L3	CO3
(OR)					
6.	a)	Write an algorithm to push and pop an element from linked stack.	5 M	L2	CO3
	b)	Design an algorithm to reverse a list of elements using a stack	5 M	L4	CO3
UNIT – IV					
7.	a)	Explain the concept of representing Queue using Single Linked List and demonstrate the basic operations of Queue.	5 M	L2	CO4
	b)	Explain the applications of a queue with appropriate examples	5 M	L2	CO4
(OR)					
8.	a)	Outline the concept of Queue and write an algorithm for Enqueue(insert) and Dequeue(delete) operations.	5 M	L2	CO4
	b)	Develop an algorithm to display all the elements of a queue	5 M	L3	CO4
UNIT – V					
9.	a)	Explain three standard ways of traversing a binary tree T with a recursive algorithms	5 M	L2	CO5
	b)	Briefly describe hashing techniques with its collision resolution strategies.	5 M	L2	CO5
(OR)					
10.	a)	What is a Binary Search Tree (BST)? Show the structure of the binary search tree after adding each of the following values in that order: 10, 25, 2, 4, 7, 13, 11, 22.	5 M	L3	CO5
	b)	A hash table contains 7 buckets and uses linear probing to solve collision. The key values are integers and the hash function used is key%7. Draw the table that results after inserting in the given order the following values: 16,8,4,13,29,11,22.	5 M	L3	CO5

***** End of the Question Paper *****